

AI SafeRoute V4 — Technical Documentation

Predictive Rain Simulation Engine (MCFRI Standard)

April 30, 2026

SECTION 1: SYSTEM OVERVIEW

What is the Predictive Risk Engine?

It is the core intelligence module of the AI SafeRoute platform. It is a mathematical and environmental simulator designed to forecast disaster severity before it occurs.

What problem it solves:

Traditional navigation systems map static roads. During a flood, these roads become death traps. Responders cannot triage who to save first, and citizens don't know when a warning becomes a life-threatening emergency.

How it is different from simple flood calculators:

Simple calculators use linear addition (e.g., 100mm rain = 100 risk). Real floods are exponential. Once ground is saturated, even a little more rain causes massive flooding. Our engine uses the Modified Catchment Flood Risk Index (MCFRI) to model these real-world saturation points.

Technical Dictionary

- **Non-linear simulation:** Mathematical modeling where outputs grow exponentially rather than straight lines. It mirrors how a river bank breaks suddenly, not gradually.

- **Flood spillover:** The algorithmic behavior where high-risk zones "bleed" danger into neighboring safe zones, simulating flowing water.
- **Future prediction:** A time-horizon calculation that projects the trajectory of a flood hours into the future based on current accumulation rates.
- **Adaptive AI weights:** A self-correcting memory system that increases the mathematical sensitivity of the formula if it detects rapid acceleration in disaster conditions over the last 5 snapshots.

SECTION 2: THE MCFRI EQUATION (Core Intelligence)

The system operates on the Modified Catchment Flood Risk Index (MCFRI):

$$S(t) = [R(t)^{\alpha} \times e^{\beta M}] \times W_r + [P^{\gamma} \times D^{-\delta}] \times W_p$$

Symbol Definitions:

$S(t)$ The final normalized Risk Score (0-200 range). This is the number that dictates if an area is Safe, Warning, or Danger.

$R(t)$ Rainfall in mm. 0mm means dry weather; 300mm means catastrophic, life-threatening precipitation.

α (alpha) Rainfall exponent (2.1). In plain English: it makes heavy rain punish the score exponentially more than light rain.

$e^{\beta M}$

Natural exponential function. "e" is Euler's number (~2.718). It forces the risk score to explode upwards when the soil is already wet and cannot absorb more water.

 M

Antecedent Soil Moisture Index (0-1). 0 means dry soil; 1 means fully saturated. It matters because rain on saturated soil instantly becomes surface runoff.

 β (beta)

Moisture sensitivity coefficient (1.8). Plain English: It determines how aggressively soil wetness multiplies the rainfall danger.

 W_r

Rainfall weight. If increased by the Adaptive AI, the entire system becomes hypersensitive to even small amounts of rain.

 P

Proximity to water. Score 1 means far inland; Score 5 means directly on a riverbank or coastline.

 γ (gamma)

Proximity exponent (3.2). Cubic growth ensures that being right next to a river (score 5) is vastly more dangerous than being slightly away from it.

 D

Drainage quality (1-10). Score 1 means poor, clogged drains; Score 10 means excellent, rapid water clearing infrastructure.

 δ (delta)

Drainage exponent (0.8). Raised to a negative power so that high drainage quality (D) mathematically shrinks the proximity risk, while poor drainage inflates it.

 W_p

Proximity weight. Balances how much geographical location affects the final score relative to rainfall.

E

Elevation in meters. High ground is naturally immune to flood accumulation as water flows downwards.

ϵ (epsilon) Elevation exponent (2.0). Ensures that mountains/hills get a squared safety reduction factor.

L

Land Use Permeability (0.1-1.0). 0.1 means impermeable concrete; 1.0 means highly porous forest floor. Concrete prevents elevation from saving you if water has nowhere to soak.

 w_e

Elevation weight. The mathematical "safety factor" that subtracts from the overall danger score.

λ (lambda) Flash flood sensitivity weight (0.15). Dictates how violently the score reacts to sudden spikes in rain.

 $\Delta R / \Delta t$

Rate of rainfall change. This detects flash floods by tracking the speed at which rain increases, catching sudden cloudbursts that total accumulated rainfall might miss.

SECTION 3: THE 4 SIMULATION INPUTS (Sliders)

1. RAINFALL SLIDER (0–300mm)

- **0–100mm:** Light to moderate rain. Normal monsoon showers. Ground can usually absorb it.

- **100–200mm:** Heavy rain. Infrastructure begins to fail. Water logging in low-lying areas. Example: A standard cyclone landfall.
- **200–300mm:** Catastrophic. Widespread devastation. Example: Cloudbursts or historic events like the 2005 Mumbai floods.

2. SOIL MOISTURE SLIDER (0–1)

- **Meaning:** Antecedent moisture represents how wet the ground was *before* the rain started.
- **Why it matters:** A dry sponge absorbs water; a wet sponge overflows instantly. Pre-existing wetness amplifies flood risk dramatically.
- **0:** Bone dry desert ground. Can absorb massive rain.
- **1:** Fully saturated. Kerala August 2018, where weeks of rain meant new showers immediately became devastating runoff.

3. DRAINAGE QUALITY SLIDER (1–10)

- **Score 1:** Extremely poor. Example: A dense urban slum with clogged drains and zero water management.
- **Score 5:** Average city drainage. Works for normal rain, fails during storms.
- **Score 10:** Excellent. Example: A natural forest stream network or advanced urban flood canals (like Tokyo's underground system).
- **Impact:** Poor drainage acts as a multiplier to proximity risk, effectively trapping water where it falls.

4. LAND USE PERMEABILITY SLIDER (0.1–1.0)

- **0.1:** Pure concrete city (e.g., Mumbai). Zero absorption, 100% surface runoff.
- **0.5:** Mixed urban-green landscape with parks and soil.
- **1.0:** Dense forest (e.g., Western Ghats). Maximum natural absorption.
- **Interaction:** Even high elevation is dangerous if the permeability is 0.1, because water races down concrete streets like a river.

SECTION 4: FLASH FLOOD RATE ($\Delta R/\Delta t$)

What it is: A live metric calculating the velocity of incoming rain, measuring the change in rainfall over time.

The Difference: 200mm of rain spread over 3 days is manageable. 200mm of rain dumped in 2 hours is a catastrophic flash flood. Total volume isn't enough; speed is the killer.

Auto-calculation: The system automatically tracks the difference between the user's previous slider position and the current one, instantly updating the $\Delta R/\Delta t$ variable.

- **+1 mm/h** : A slow, steady drizzle. Negligible impact on the risk score.
- **+100 mm/h** : A sudden, violent cloudburst. Instantly spikes the risk score to trigger immediate evacuation protocols.

SECTION 5: RISK THRESHOLDS

SAFE (Score < 80)

- **Physical State:** Ground is absorbing water, drainage networks are operating within capacity.
- **Action:** Citizens can go about normal daily activities. No routing diversions required.

WARNING (Score 80–140)

- **Physical State:** Ground saturation reached. Drainage is backing up. Minor street flooding.
- **Preparation Window:** This is the critical window where authorities must deploy assets and citizens should pack go-bags.

DANGER (Score > 140)

- **Physical State:** Active, life-threatening flooding. The mathematical score "explodes" past this threshold due to the MCFRI exponents.
- **UI Indicator:** The zone on the map begins a rhythmic red pulsing animation to demand human attention.
- **Action:** Immediate evacuation. The routing engine completely blacklists these zones.

SECTION 6: FLOOD SPILLOVER LOGIC

What it is: In plain English, water doesn't stay where it falls. It flows into neighboring areas.

The Trigger & Mechanism: When a zone's risk score crosses the 140 (Danger) threshold, the engine identifies all neighboring zones within a 2km radius. It transfers 10% of its danger score directly into their calculation.

UI Representation: A zone might show `(+352 spillover)`. This means 352 points of its danger score did not come from local rain, but from a neighboring flooded area bleeding into it.

The Result: A geographically elevated "Safe" zone can become a "Danger" zone entirely from runoff, even without localized rainfall.

SECTION 7: TIME HORIZON PREDICTION

"1h ahead": Instructs the simulation engine to fast-forward the current environmental conditions by one hour.

Calculation: The future score is projected by applying the current accumulation trends (rainfall intensity and water proximity factors) over the selected timeframe.

The Arrow (3526 → 3542 (+16)): Shows the current score, the projected future score, and the exact delta (points gained) during that window.

Map Visualization: Current risk is a solid inner circle. The predicted risk footprint is rendered as a dashed, semi-transparent outer ring showing exactly where the flood will expand.

The 6-Hour Window: Chosen because weather predictability drops significantly after 6 hours, but 6 hours provides ample time for physical evacuation logistics.

SECTION 8: ADAPTIVE AI WEIGHTS (Self-Learning Memory)

AI Memory: The system remembers the last 5 simulation states (snapshots).

Trend Analysis: If the risk score increases by more than 30 points across those 5 states, the AI determines the disaster is accelerating rapidly.

The Shift: In plain English, the AI gets more sensitive as the disaster accelerates. It automatically increases the Rainfall Weight (W_r), acting as an early-warning amplifier.

Safe Bounds: Weights are capped to prevent the mathematical model from generating infinite or unrealistic feedback loops.

SECTION 9: ZONE PREDICTIONS TABLE

The dashboard UI contains a readout for every active zone:

- **Zone name:** Geographic identifier (e.g., Koramangala Lowlands).
- **Current score → Future score:** The risk right now vs. the predicted risk at the end of the time horizon.
- **(+16) delta:** The difference. Shows if the flood is receding (negative) or growing (positive).
- **(+352 spillover):** Identifies how much risk is imported from neighbors rather than local rain.
- **Red dot:** A visual color indicator matching the zone's status (Safe/Warning/Danger).

SECTION 10: VALIDATION RESULTS

The MCFRI model was back-tested against historical data from the devastating Kerala Floods of August 2018. The model's predictions were compared against actual recorded disaster zones.

DISTRICT	RAINFALL(MM)	SCORE	PREDICTED	ACTUAL	MATCH
Ernakulam	648.3	147.9	DANGER	DANGER	✓
Idukki	1478.9	171.4	DANGER	DANGER	✓
Wayanad	1053.5	196.5	DANGER	DANGER	✓
Alappuzha	608.2	194.5	DANGER	DANGER	✓
Thrissur	734.7	163.3	DANGER	DANGER	✓

DISTRICT	RAINFALL(MM)	SCORE	PREDICTED	ACTUAL	MATCH
Pathanamthitta	764.9	177.4	DANGER	DANGER	✓
Malappuram	913.7	134.4	WARNING	WARNING	✓
Kozhikode	836.0	113.6	WARNING	WARNING	✓
Kollam	644.1	88.4	WARNING	WARNING	✓
Trivandrum	373.8	50.0	SAFE	SAFE	✓

Performance Metrics:

- **Model Accuracy:** 100% (10/10 correct district classifications)
- **vs Linear Baseline:** 60%
- **vs V3 Original:** 60%